MOTION OF A CHARGED PARTICLE IN A MAGNETIC FIELD

In previous sections we studied about sources of magnetic field. In this section we shall study the effect of magnetic field on a moving charge and on current carrying wire.

In magnetic field force experienced by a charged particle is given by the expression.

$$\vec{F}_m = q(\vec{v} \times \vec{B})$$

Here, $\vec{v}$ = velocity of the particle and $\vec{B}$ = magnetic field

On the basis of above expression, we can draw following conclusions.

- Stationary charge (i.e. $\vec{v} = 0$) experiences no magnetic force.
- If $\vec{v}$ is parallel or anti parallel to $\vec{B}$ then the charged particle experiences no magnetic force.
- Magnetic force is always perpendicular to both $\vec{v}$ and $\vec{B}$.
- As magnetic force is always perpendicular to $\vec{v}$, it does not deliver any power to the charged particle.
- As magnetic force is always perpendicular to the $\vec{v}$, it will only cause a change in the direction of $\vec{v}$.
- On the basis of expression $\vec{F} = q(\vec{v} \times \vec{B})$ the maximum value of magnetic force is equal to F = qvB, which occurs when the charge is projected perpendicular to the uniform magnetic field. In this case path of charged particle is circular and magnetic force provides the necessary centripetal force.
- If radius of the circular path is R then,

$$\frac{mv^2}{R} = qvB, \text{ where, } m = \text{mass of the particle}$$

$$\Rightarrow \quad R = \frac{mv}{qB}$$

- Time taken to complete one revolution is

$$T = \frac{2\pi R}{v}$$

$$T = \frac{2\pi m}{qB}$$

If the charged particle is projected obliquely to the field then its velocity $\vec{V}$ can be resolved into two components, one along the $\vec{B}$ say $V_{\parallel}$ and the other perpendicular to $\vec{B}$ say $V_{\perp}$. It experiences a magnetic force and hence has a tendency to move on a circular path. Due to $V_{\parallel}$ it experience no force, and hence has a tendency to move on a straight path along the field. So in this case it moves along a helical path.

- Radius of the helix is $R = \frac{mV_{\perp}}{qB}$
- Time taken to complete one revolution is $T = \frac{2\pi m}{qB}$

  (Note: T is independent of V)
- The distances moved by the charged particle along the magnetic field during one revolution is called pitch.

$$\Rightarrow \quad \text{pitch} = V_{\parallel} \times T = \frac{2\pi m v_{\parallel}}{qB}$$

- Force exerted by constant magn. field $\vec{B}$ on conductor of length $\vec{l}$ carrying current i is $\vec{F} = i\vec{l} \times \vec{B}$.

- $\vec{l}$ = vector length / shortest distance betn end points of the conductor.

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- if $\vec{l} = \vec{0}$, then $\vec{F}_{mag} = i\vec{l} \times \vec{B} = \vec{0}$ i.e no force is experienced by a closed current carrying loop in a uniform magnetic field.

- Electromagnetic force on a charge 'q' moving with velocity $\vec{v}$ in both electric & magnetic fields $\vec{E}$ & $\vec{B}$ is

$$\vec{F} = \vec{F}_{el} + \vec{F}_{mag}$$

$$= q\vec{E} + q(\vec{v} \times \vec{B})$$

<u>Note</u> $\vec{F}_{el} = q\vec{E}$ performs work & changes KE & speed of charge

but $\vec{F}_{mag} = q(\vec{v} \times \vec{B})$ is no work force & hence it can not change KE & speed of charge. It can only change dir. of velocity vector $\vec{v}$.

- If $\vec{E} \perp \vec{B}$ & $\vec{v}$ of a charge q is also $\perp$ to both $\vec{E}$ & $\vec{B}$ & if the charge 'q' moves with constant velocity through these $\perp$ $\vec{E}$ & $\vec{B}$ then $\boxed{v = \frac{E}{B}}$

- If charge 'q' passes through a p.d 'V' & gains a velocity 'v' from rest then for work energy theorem,

$$W = \Delta KE$$

$$qV = \frac{1}{2}mv^2$$

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Laplace's Law, Biot Savart's Law, Ampere's Law & their applications are not required to solve problem of Magnetism Set -1.

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